Laxmi Charitable Trust's

Sheth L.U.J. College of Arts & Sir M.V. College of Science & Commerce

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Department of Information Technology (Bsc.IT Semester III)

Python Programming Practical

Practical – 6

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Class :	Date/Time of Submission: 28-07-2026

Practical No. 6 - Write Algorithm, Code and Output

a. Write a program to compute number of characters and words in a string.

Code:

prac6.py - C:/Users/mvluc/Desktop/prac6.py (3.14.6)

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```
#Write a program to compute number of characters and
#words in a string.
'''Use split() method for word count. It divides the
string into words using whitespaces as the delimiter
and len() counts the number of elements in the
resulting list.len(s) counts the total number of
characters in the string
(including letters,spaces and punctuation)'''
s="Second Year IT students"
word_count=len(s.split())
Char_count=len(s)

#Printing Result
print("The number of words in a string are :"+str(word_count))
print("The number of characters in a string are :", Char_count)
print("laxmi jha S013")
```

Output:

```
>>>
=================== RESTART: C:/Users/mvluc/Desktop/prac6.
The number of words in a string are :4
The number of characters in a string are : 23
laxmi jha S013
>>>
```

b. Create a file geometry.py to calculate base areas for shapes square and circle. In another file, write a function pointyShapeVolume(x, y, squareBase) that calculates the volume of a square pyramid if squareBase is True and of a right circular cone if squareBase is False. x is the length of an edge on a square if squareBase is True and the radius of a circle when squareBase is False. y is the height of the object. First use squareBase to distinguish the cases. Use the circleArea and squareArea from the geometry module to calculate the base areas.

Code:

prac6.py - C:/Users/mvluc/Desktop/prac6.

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```
import math

def circleArea(r):
    return math.pi * r ** 2

def squareArea(x):
    return x ** 2
```

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```python
import geometry

def pointyShapeVolume(x, y, squarebase):

    if squarebase:
        base_area = geometry.squareArea(x)    # Volume of square or rectangular pyramid

    else:
        base_area = geometry.circleArea(x)    # Volume of a cone

    return y * base_area / 3.0
print("Volume (Square Base):", pointyShapeVolume(4, 2.6, True))
print("Volume (Circular Cone):", pointyShapeVolume(4, 2.6, False))
```

output:

```
>>>
=================== RESTART: C:/Users/mvluc/Desktop/prac6.py
Volume (Square Base): 13.866666666666667
Volume (Circular Cone): 43.56341812977846
>>>
```

Q. 2. How can you insert a substring into a string?

Q. 3. List various functions of string with examples.